MECH 491/542 Assignment #3

Due: **February 28th, 11:59 pm** on Canvas under Assignment #3 (late penalty – 20% per day)

QUESTION (10 marks):

Christian Rowsell (40131393)

Complete the following word address program (G-code) to finish machine the component shown in the figure below:

N0010 G20 G17 G40 G49 G80 G98 G90 G00 N0020 G53
Z0.

N0030 T02 M06

N0040 G00 G54 X-1.0. Y-1.0 M03 S1200

.

.

.

.

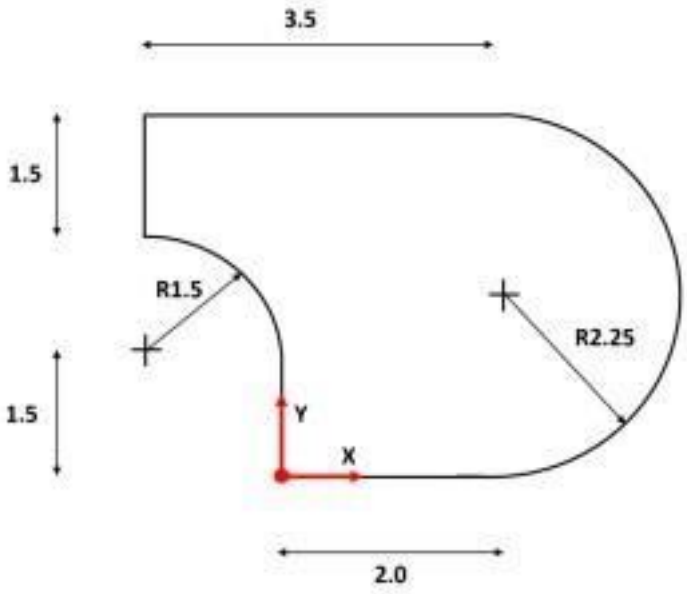
.

N00… M30

T02: ½" diameter endmill is used

Part Zero: lower left corner at the top of the work part (shown in red) Part (1 inch

thick):



Units: Inch.

2022W T2

The following G-code can be used to create the given part. It is assumed that any text coming after a semi-colon is a comment. These comments will be used to help explain each line. It is assumed that no coolant will be used for this part, and that the tool diameter and height will be given in the program as D2 and H2 respectively. It is assumed that a feed rate of 15in/min has been previously calculated for this program.

N0010 G20 G17 G40 G49 G80 G98 G90 G00; Set imperial, set X-Y plane, cancel cutter offset, cancel tool length offset, cancel fixed cycle, cancel cycle return point, absolute movement, rapid movement

N0020 G53 Z0; Set machine coordinate, set Z to 0

N0030 T02 M06; Select tool 02, change tool

N0040 G00 G54 X−1.0 Y-1.0 M03 S1200; Rapid movement, set machine coordinate, set x -1, set y – 1, spindle on, 1200 rpm

N0050 G41 D2; Set cutter offset right by diameter of tool 2

N0060 G43 H3; Set cutter vertical offset by height of tool 2

N0070 G00 X0.0 Y-1.0; Align cutter with X axis, move cutter below part

N0080 G00 Z1; Set cutter to sit 1 inch above material

N0090 G01 Z-1.0 F10.; Move tool to 1 inch below surface, set feed rate to 10in/min

N0100 Y1.5 F15.; Move tool to beginning of first curve.

N0110 G03 X-1.5 Y3.0 I-1.5 J0.0; Counter clockwise turn, final position X-1.5 Y3 Center point -1.5 in X from cutter starting point

N0120 G01 Y4.5; Move cutter to top left corner of part

N0130 G01 X2.0; Move cutter to beginning of second radius. (The coordinates are absolute, X starts at −1.5 and will move to 2.0 for the total distance of 3.5 found in drawing.)

N0140 G02 X2.0 Y0.0 I0.0 J-2.25; Clockwise turn, final position X2 Y0, center point -2.25 (1 radius) in Y from cutter starting point

N0150 G01 X0.0; Move cutter back to initial point

N0160 X-1.0 Y-1.0; Move cutter away from part

N0170 G00 Z1.0; Move cutter above part

N0180 X5.0 Y5.0; Move cutter away from part to allow for part removal

N0190 M05; Turn off cutter

N0200 G40; Turn off cutter compensation

N0210 M30; Tell machine that program is finished.

The same code can be found without the comments below.

N0010 G20 G17 G40 G49 G80 G98 G90 G00;

N0020 G53 Z0;

N0030 T02 M06;

N0040 G00 G54 X−1.0 Y-1.0 M03 S1200;

N0050 G41 D2;

N0060 G43 H3;

N0070 G00 X0.0 Y-1.0;

N0080 G00 Z1;

N0090 G01 Z-1.0 F10.;

N0100 Y1.5 F15.;

N0110 G03 X-1.5 Y3.0 I-1.5 J0.0;

N0120 G01 Y4.5;

N0130 G01 X2.0;

N0140 G02 X2.0 Y0.0 I0.0 J-2.25;

N0150 G01 X0.0;

N0160 X-1.0 Y-1.0;

N0170 G00 Z1.0;

N0180 X5.0 Y5.0;

N0190 M05;

N0200 G40;

N0210 M30;

A tool to simulate G-Code movement was used to validate this code. The following shape was created using this G-code.

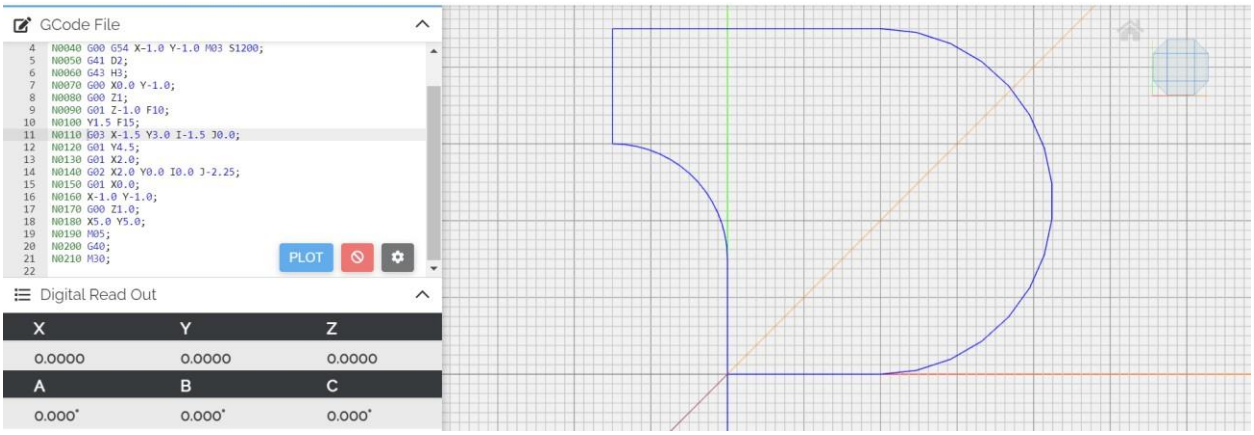


Figure 1: G-Code Trace [1]

Sources used:

[1] - https://ncviewer.com/ https://www.thomasnet.com/articles/custom-manufacturing-fabricating/g-code-commands/ https://www.cnccookbook.com/g-code-m-code-command-list-cnc-mills/ The class notes